COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 Data Structures and Algorithms	**SECTION:**	CPE21S1
DATE PERFORMED:	7/28/2026	**DATE SUBMITTED:**	7/31/2026
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ACTIVITY NO. _4__ STACKS

1. PROCEDURE OUTPUT

ILO A

main.cpp

```cpp
#include<iostream>
#include<stack> // calling stcak from the STL

int main(){
    // create an object for the stack
    std::stack<int> newStack;

    std::cout << "Stack Empty? " << newStack.empty() << std::endl;
    newStack.push(3);
    newStack.push(8);
    newStack.push(15);

    std::cout << "Stack Empty? " << newStack.empty() << std::endl;
    std::cout << "The size of the stack: " << newStack.size() << std::endl;
    std::cout << "The top of the element is: " << newStack.top() << std::endl;
    std::cout << "Removing elements from the stack: \n";
    newStack.pop();
    std::cout << "The top of the element is: " << newStack.top() << std::endl;
    newStack.pop();
    std::cout << "The top of the element is: " << newStack.top() << std::endl;
    newStack.pop();
    std::cout << "Stack Empty? " << newStack.empty() << std::endl;
```

output

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

mz.h0n' '--stderr=Microsoft-MIEngine-Error-adlrem5w.ww4' '-
Microsoft-MIEngine-Pid-ydwxnt4q.pte' '--dbgExe=C:\msys64\uc
bin\gdb.exe' '--interpreter=mi'
Stack Empty? 1
Stack Empty? 0
The size of the stack: 3
The top of the element is: 15
Removing elements from the stack:
The top of the element is: 8
The top of the element is: 3
The top of the element is:
```

ILO B

```
int main(){
    std::cout << "Enter the number of max elements for the new stacks: ";
    std::cin >> i;
    while(true){
        std::cout << "Stack operation" << std::endl;
        std::cout << "1. Push|2.Pop|3.Top|4.isEmpty|5.Display all" << std::endl;
        std::cin >> choice;
        switch(choice){
            case 1: push();
            break;
            case 2: pop();
            break;
            case 3: Top();
            break;
            case 4: std::cout << isEmpty() << std::endl;
            break;
            case 5: displayALL();
            break;
            default: std::cout << "invalid choice." << std::endl;
        }
    }
    return 0;
}
```

```
79  void displayALL(){
80      if (isEmpty()){
81          std::cout << "No stack to display" << std::endl;
82      } else {
83          for(int i = 0; i != choice; i++){
84          Top();
85          pop();
86          }
87      }
88      std::cout << "All stack has been display and remove" << std::endl;
89  }
```

THIS IS THE CODE THAT I MODIFIED ON ILO A

output

```
Enter the number of max elements for the new stacks: 5
Stack operation
1. Push|2.Pop|3.Top|4.isEmpty|5.Display all
1
New Value to add into the stack
30
Stack operation
1. Push|2.Pop|3.Top|4.isEmpty|5.Display all
1
New Value to add into the stack
60
Stack operation
1. Push|2.Pop|3.Top|4.isEmpty|5.Display all
1
New Value to add into the stack
80
Stack operation
1. Push|2.Pop|3.Top|4.isEmpty|5.Display all
1
New Value to add into the stack
100
```

```
Stack operation
1. Push|2.Pop|3.Top|4.isEmpty|5.Display all
1
New Value to add into the stack
1
Stack operation
1. Push|2.Pop|3.Top|4.isEmpty|5.Display all
1
New Value to add into the stack
555
Stack operation
1. Push|2.Pop|3.Top|4.isEmpty|5.Display all
1
Stack overflowStack operation
1. Push|2.Pop|3.Top|4.isEmpty|5.Display all
3
The element on the top of the stack is: 555
```

```
Stack operation
1. Push|2.Pop|3.Top|4.isEmpty|5.Display all
5
The element on the top of the stack is: 555
Popping: 555
The element on the top of the stack is: 1
Popping: 1
The element on the top of the stack is: 100
Popping: 100
The element on the top of the stack is: 80
Popping: 80
The element on the top of the stack is: 60
Popping: 60
All stack has been display and remove
Stack operation
1. Push|2.Pop|3.Top|4.isEmpty|5.Display all
```

Conclusion

- In the ILO B, i tried to add a new stack like this stack2[maxCap], the purpose is that i dont know how to display the next top if i didnt pop the top. So i figure that i will make a new stack2[maxCap], with this i can display and pop the data inside the stack[maxCap] without deleting permanently the data inside the stack. stack[maxCap] == stack2[maxCap], so i did this but it doesnt work so i delete the idea of making a new stack and proceed to displaying it and removing the top so i can display the new top.
- To execute that, i use the for loop function to display the top and pop it, the results of that is the output above.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

ILO C Header file

```
#ifndef STACK_LIST_H
#define STACK_LIST_H

// class node that contains data and the next
template <typename T>
class Node{
    public:
    T data;     // Value that we need store
    Node* next; // pointer to the next node
};


// global pointer for the top of the stack
template <typename T>
Node<T> *head = nullptr;


// push
template <typename T>
void push(T newData){
    // create a new node and assign the newData
    Node<T>* newNode = new Node<T>;
    newNode->data = newData;
```

```
24
25      // new node points to current top, then becomes the new top
26      newNode->next = head<T>;
27      head<T> = newNode;
28  }
29  // pop
30  template <typename T>
31  T pop(){
32      // check if the stack is empty
33      if (head<T> == nullptr){
34          std::cout << "stack underflow" << std::endl;
35      }
36      // save the current top node
37      Node<T>* temp = head<T>;
38      //store its data
39      T tempVal = temp->data;
40      // move the head to the next node
41      head<T> = head<T>->next;
42      // free the removed node
43      delete temp;
44      // return the removed value
45      return tempVal;
46  }
```

```
49  // isEmpty
50  template <typename T>
51  bool isEmpty(){
52      return head<T> == nullptr;
53  }
54
55  // top
56  template <typename T>
57  void Top(){
58      // error catching
59      if(head<T> == nullptr){
60          std::cout << "Stack Underflow" << std::endl;
61      }
62      // display the top node
63      std::cout << "The top of the stack is: " << head<T>->data << std::endl;
64  }
```

```cpp
// display all
template <typename T>
void displayAll(){
    if (head<T> == nullptr){
        std::cout << "stack underflow\n";
        return;
    }
    std::cout << "Displaying the stack:\n";
    Node<T>* temp = head<T>;
    while(temp != nullptr){
        std::cout << temp->data << " ";
        temp = temp->next;
    }
    std::cout << std::endl;
}

template <typename T>
void checking(){
    T input;
    std::cout << "Enter the expression ('~' to stop): " << std::endl;
    while(std::cin >> input && input != '~'){
        if (input != '(' && input != '{' && input != '[' && input != ')' && input != '}' && input != ']') {
            std::cout << "Ignore: " << input << std::endl;
        }
        else if (input == '(' || input == '{' || input == '[') {
            push(input);
            std::cout << "Pushing: " << input << std::endl;
        }
        else if (input == ')' || input == '}' || input == ']') {
            if (head<T> == nullptr) {
                std::cout << "Error: Stack Underflow." << input << std::endl;
                return;
            }
            else {
                T popped = pop<T>();
                if ((input == ')' && popped != '(') ||
                    (input == '}' && popped != '{') ||
                    (input == ']' && popped != '[')) {
                    std::cout << "Error: Mismatched symbols! Popped " << popped << " but found " << input << std::endl;
                    return;
                } else {
                    std::cout << "Matched: " << input << std::endl;
                }
            }
        }

    }
    if (head<T> != nullptr) {
        std::cout << "There is an error on the line.\n";
    } else {
        std::cout << "Success!\n";
    }
}
```

ILO C Main cpp

main-2.cpp | stacklist.h

qwe > main-2.cpp > main()

```cpp
#include<iostream>
#include "stacklist.h"

int main(){
    std::cout << "----- Pushing 'c' 'p' 'e' Onto the stack ----\n";
    // push
    push('c');
    push('p');
    push('e');

    // Display all elements currently in the stack
    displayAll<char>();

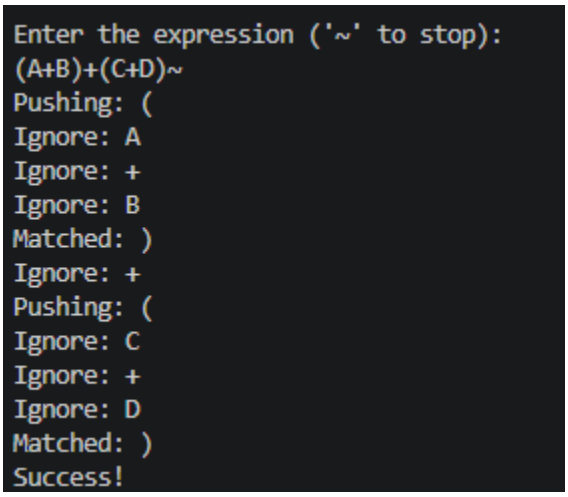
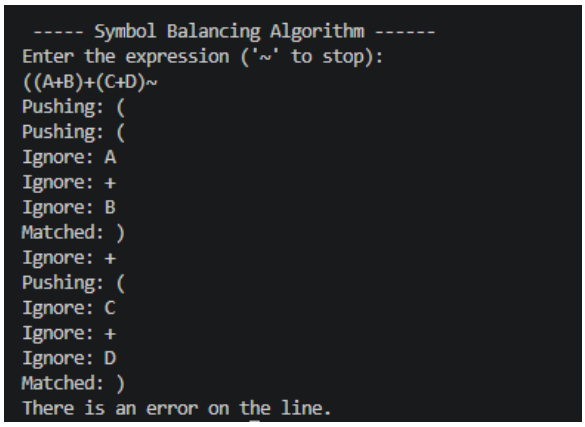
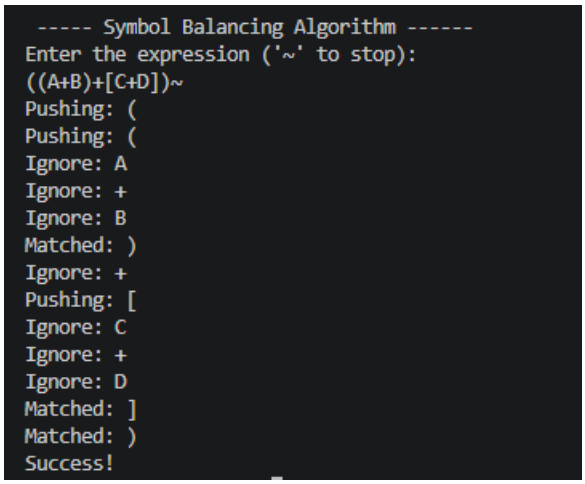
    // display the current top
    Top<char>(); // Top should be 'e'

    std::cout << "\n ----- Popping all the elements -----\n";

    while(!isEmpty<char>()){
        std::cout << "Popped: " << pop<char>() << std::endl;
    }
    std::cout << "\n ----- Testing the Underflow ------\n";
    pop<char>();

    std::cout << "\n ----- Symbol Balancing Algorithm ------\n";
    checking<char>();

    return 0;
}
```

Expression	Valid Y/N	Output	Analysis
(A+B)+(C+D)	Y		It is true because the opening and closing delimiter is matched
((A+B)+(C+D)	N		It is false because there is a one closing delimiter left, it means it is unmatched to the opening delimiter
((A+B)+[C+D])	Y		It is true because the opening and closing delimiter is matched

((A+B]+[C+D]}	N	----- Symbol Balancing Algorithm ------ Enter the expression ('~' to stop): ((A+B]+[C+D]}~ Pushing: (Pushing: (Ignore: A Ignore: + Ignore: B Error: Mismatched symbols! Popped (but found]	It is false because the opening and closing delimiter is unmatched, the other one is (while the other one is]

OUPUT

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
ne-Error-aodg2rk4.fo5' '--pid=Microsoft-MIEngine-Pid-aeovkuez.jwe'
----- Pushing 'c' 'p' 'e' Onto the stack ----
Displaying the stack:
e p c
The top of the stack is: e

 ----- Popping all the elements -----
Popped: e
Popped: p
Popped: c

 ----- Testing the Underflow ------
stack underflow
```

```
 ----- Symbol Balancing Algorithm ------
Enter the expression ('~' to stop):
(A+B)+(C+D)~
Pushing: (
Ignore: A
Ignore: +
Ignore: B
Matched: )
Ignore: +
Pushing: (
Ignore: C
Ignore: +
Ignore: D
Matched: )
Success!
PS C:\Emptech\IGLOSO>
```

Conclusion

- By working on this laboratory, I have gained a lot of knowledge regarding stacks and their workings with the help of linked list in C++. Working on the expression balancing algorithm and applying it to various expressions has given me an idea of how the compiler checks for syntactic errors and balanced brackets. Despite the fact that I have faced some errors and terminal related problems in the process, solving these issues has helped me get a better understanding of the code.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Google**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Concept of code and code assistance**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I use the google to search for the code that i forgot, like the else if and the while because im confuse if it is python or C++. and for the concept of code, i ask for the explanation to understand the problem definition.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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